Parth Nitesh Chaudhari

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Education

International Institute of Information Technology — Hyderabad, TG
Bachelor of Technology in Computer Science Engineering — *July 2024 – May 2028*

Asian International School — Abu Dhabi, UAE
High School Diploma — *Oct. 2014 – May 2024*

Experience

Undergraduate Research Assistant — April 2026 – Present
Under guidance of Professor Kamalakar Karlapalem — *Data Sciences and Analytics Center, IIIT-H*

- Developed Computer Vision software (Python, OpenCV, YOLO) that tracks passes, formations and attacking patterns in football matches, separating teams and the ball with K-Means clustering and scoring each action to visualize its impact on chances created.

Undergraduate Intern — January 2026 – May 2026
Part of DASS course — *Cosmic 3D, Maker's Lab*

- Developed a comprehensive web-based system for remote control and monitoring of 3D printers, enabling internet-based operation from any browser.
- Implemented real-time MQTT communication for instant updates between backend API, React dashboard, and ESP32 firmware, supporting live telemetry and hardware control.
- Integrated file management with G-code slicing, 3D model visualization using Three.js, and authentication for secure multi-user access to printer networks.

Projects

LLM Confidence Under Pressure | *PyTorch, Hugging Face, W&B* — September 2026 – October 2026

- ANLP course project: built a reproducible multi-turn pipeline that puts Qwen2.5-7B and Llama-3.1-8B through six turns of criticism, praise or a neutral control on 450 claims of known truth (MATH-500, real analysis), logging verdicts, token-probability probes and stated confidence per turn.
- Showed that criticism cuts Llama's retention of correct MATH answers from 68.6% to 14.7% (Qwen: 93.9% to 76.3%) and its true/false discrimination from 0.46 to 0.11; harsher tone only mattered when paired with a claim that the answer was wrong. When Qwen gave in, its own working still reached the right answer 73% of the time.

C-Shell | *C, POSIX* — September 2025

- Developed a POSIX-compliant UNIX shell in C featuring a CFG command parser, multi-stage pipelines, and I/O redirection. Implemented custom signal handling (SIGINT/SIGTSTP), job control (fg/bg), asynchronous process reaping, and persistent session history.

Network Protocol Suite | *C, HTTP, TCP, UDP, DNS* — September 2025 – October 2025

- Developed a custom C TCP-like protocol over UDP featuring handshakes, sliding-window flow control, cumulative ACKs, and retransmissions. Built client-server chat and file transfer applications (with MD5 validation) utilizing select() for non-blocking I/O multiplexing.
- Developed a terminal network sniffer in C using libpcap and Berkeley Packet Filters (BPF). The tool captures network frames in real-time, decodes protocols across the TCP/IP stack (Ethernet, IPv4/IPv6, ARP, TCP/UDP), and provides an in-memory session analyzer for offline Hex/ASCII payload forensics.

CLI - Docs | *C, TCP Sockets, POSIX* — September 2025

- Developed a distributed shared-document system in C using TCP sockets and a multi-threaded naming/storage server architecture. Engineered sentence-level lock managers for concurrent, conflict-free editing, and implemented owner-approvable access requests. Built metadata persistence, word-by-word content streaming, and a universal error-handling framework.

Technical Skills

Languages: Python, C/C++, SQL, JavaScript, HTML/CSS, Arduino C++
Frameworks: PyTorch, Hugging Face Transformers, TensorFlow, Scikit-Learn, React, Node.js, Express, FastAPI
Developer Tools: Git, Docker, Google Cloud Platform, Weights & Biases, Slurm, Linux, VS Code, PyCharm, ESP-32, MongoDB
Libraries: pandas, NumPy, Matplotlib, OpenCV, YOLO, bitsandbytes